



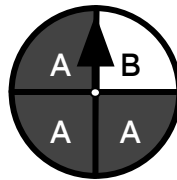
Problem solving with possible outcomes

Task A: Outcome tables and likelihood

1) Alex will spin this spinner twice.

a) Complete this outcome table to list all unique outcomes.

	A	B
A		
B		



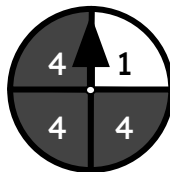
spinner 1

b) Construct your own outcome table which shows the most likely outcome.

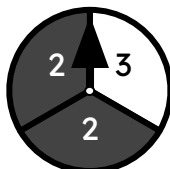
2) Aisha will spin spinner 2 and spinner 3 twice, and find the product of the numbers the spinners land on.

x	1	

a) Complete this outcome table to list all unique outcomes.



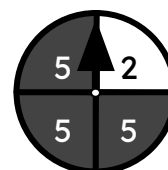
spinner 2



spinner 3

b) Construct a second outcome table. Starting with the least likely outcome, rank the outcomes in order of likelihood.

3) Sam and Lucas play a game. Sam will spin spinner 4 twice and find the sum of the numbers the spinner lands on. If the score is a double-digit number, Sam wins, if the score is either 8 or 9, it's a draw, and if the score is 7 or less, Lucas wins.



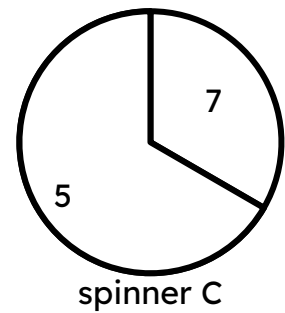
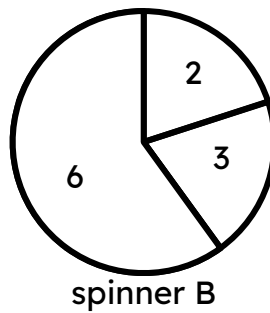
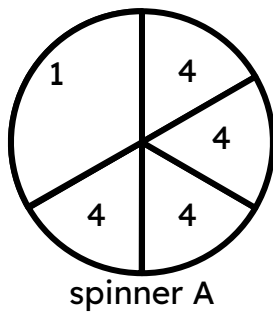
spinner 4

a) Construct the most efficient outcome table to show it is impossible for Sam and Lucas to draw.

b) Construct the most efficient outcome table to find whether Sam or Lucas are more likely to win. Explain how you know.



Task B: Outcomes and unequal likelihood



1) Jacob will spin spinners A and B once each, with the number each spinner lands on multiplied together. Construct an outcome table which lists each unique outcome to this trial.

2) Split sectors on spinners A, B, C so that all outcomes on a spinner are equally likely.

3) Jacob will spin spinners B and C once each, with the number each spinner lands on multiplied together. Construct an outcome table which shows how likely each outcome is to occur.

4) Jacob will spin spinners A and B once each, then spinners B and C once each, then spinners A and C once each. After each trial, the number each spinner lands on are multiplied together.

Which outcome is most likely to occur across all experiments?